GROUP 14
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«abstract» User
-int userId
-String username
-String passwordHash
-String fullName
-String email
-String contactNo
-String role
-String accountStatus
+boolean login(String username, String password)
+void logout()
+abstract void updateProfile()
+List<Notice> viewNotices()
#void validatePassword(String pwd) throws InvalidPasswordException

Admin
+User createUser(UserProfileDTO profile)
+void updateUser(int userId, UserProfileDTO profile)
+void deleteUser(int userId)
+Course createCourse(CourseDTO course)
+void updateCourse(int courseId, CourseDTO course)
+Notice createNotice(NoticeDTO notice)
+void updateNotice(int noticeId, NoticeDTO notice)
+Timetable createTimetable(TimetableDTO tt)
+void updateTimetable(int timetableId, TimetableDTO tt)
+void updateUserProfile(int userId, UserProfileDTO profile)

Notice
-int noticeId
-String title
-String description
-String targetRole
-Date postedAt
-int postedBy
+void postNotice()
+void updateNotice()
+void deleteNotice()

CourseMaterial
-int materialId
-String title
-String filePath
-Date uploadedAt
-int lecturerId
+void uploadMaterial()
+void downloadMaterial()

Lecturer
-int lecturerId
-String designation
-int departmentId
+void updateProfile() %% polymorphism
+void addCourseMaterial(int offeringId, MaterialDTO m)
+void updateCourseMaterial(int materialId, MaterialDTO m)
+void uploadMarks(int offeringId, int assessmentId, Map marks)
+StudentDetails viewStudentDetails(int studentId)
+EligibilityStatus viewEligibility(int studentId)
+GradeReport viewGrades(int studentId)
+AttendanceReport viewAttendance(int studentId)
+List<Medical> viewMedical(int studentId)

Department
-int departmentId
-String departmentName
+Department getDepartment(int id)
+List<Lecturer> getLecturers()
+List<Course> getCourses()

Course
-int courseId
-String courseCode
-String courseName
-int theoryCredits
-int practicalCredits
+void addCourse()
+void updateCourse()
+List<CourseOffering> getOfferings()

CourseOffering
-int offeringId
-String semester
-String academicYear
-String batch
-int lecturerId
+void enrollStudent(int studentId)
+List<Enrollment> getEnrollment()
+Timetable getTimetable()

Assessment
-int assessmentId
-String assessmentName
-String assessmentType
-double weightPercentage
-int maximumMark
-String evaluationCriteria
+void addAssessment()
+void updateAssessment()

Mark
-int markId
-int enrollmentId
-int assessmentId
-double marksObtained
-int attemptNo
+void uploadMark()
+void updateMark()
+double calculateCAMark()

Enrollment
-int enrollmentId
-int studentId
-int offeringId
-String enrollmentStatus
+void markEnrollment()
+boolean checkEligibility()

Undergraduate
-int studentId
-String registrationNo
-String batch
-String academicYear
-String studentStatus
+void updateProfile() %% polymorphism - restricted
+void updateContactDetails(String email, String contactNo)
+void updateProfilePicture(String imagePath)
+AttendanceReport viewAttendance()
+List<Medical> viewMedicals()
+List<Course> viewCourses()
+GradeReport viewGrades()
+GPAReport viewGPA()
+List<Timetable> viewTimetable()

Timetable
-int timetableId
-int offeringId
-String dayOfWeek
-Time startTime
-Time endTime
-String sessionType
-String lectureHall
+void addToTimetable()
+void updateTimetable()
+List<Timetable> getWeeklyTimetable()

AttendanceSession
-int sessionId
-int offeringId
-int timetableId
-String sessionType
-Date sessionDate
-int sessionNumber
-Date recordedAt
+void recordSession()
+AttendanceSession getSession()

AttendanceRecord
-int attendanceId
-int studentId
-int sessionId
-String status
-Date recordedAt
+void markAttendance(String status)
+void updateAttendance(String status)
+double calculateAttendancePercentage()

Medical
-int medicalId
-int studentId
-int recordedBy
-Date medicalDate
-String supportingDocumentPath
-String approvalStatus
+void submitMedical()
+void approveMedical()
+void rejectMedical()
+boolean isApproved()

TechnicalOfficer
-int toId
-int departmentId
+void updateProfile() %% polymorphism
+void recordAttendance(int sessionId, Map<int,String> records)
+void updateAttendance(int attendanceId, String status)
+Medical submitMedical(MedicalDTO m)
+void updateMedicalStatus(int medicalId, String status)
+List<Timetable> viewTimetable(int departmentId)

SemesterResult
-int semesterResultId
-int studentId
-String semester
-String academicYear
-double sgpa
-double cgpa
+double calculateSGPA()
+double calculateCGPA()
+List<CourseResult> getCourseResults()

CourseResult
-int resultId
-int enrollmentId
-double caMark
-double finalMark
-double overallMark
-boolean caEligible
-boolean overallEligible
-String gradeLetter
-double gradePoint
+double calculateCA()
+boolean checkCAEligibility()
+boolean checkOverallEligibility()
+GradeRule determineGrade()

GradeRule
-int gradeRuleId
-String gradeLetter
-double minMark
-double maxMark
-double gradePoint
+String getGrade(double mark)
+double getGradePoint(String grade)

has
"targeted to"
posts
employs
offers
maintains
"offered as"
"taught by"
uploads
for
belongs to
submits
"scheduled in"
"generates"
records
produces
"covers"
"records"
"aggregates"
"determines"